What is Transit-Time Damping (TTD) and How to Identify it From Plasma Turbulence

Working Group # 4 | Presenter: Rui Huang rui-huang@uiowa.edu

Authors: Rui Huang[1], Gregory Howes[1], Andrew McCubbin[2], Alberto Felix[1] | Affiliations: [1]University of Iowa, [2]Johns Hopkins University

Goals

- Clarify ***the meaning of "TTD"*** in our context
- Display the unique ***velocity space signature*** of TTD
- ***Identify TTD*** from plasma turbulence

Motivations

- Two different uses of the term "TTD" exist in literature.
- Provide reference for TTD identification.
- Grasp the mechanisms behind plasma turbulence damping, crucial for multiple applications.

Two Uses of the Term "TTD" in Literature

Intended Meaning Here:

- First appears in Stix (1962)
- Originates from "transit-time magnetic pumping"
- "Transit-time" denotes the duration for an ion to transit the particle heating device
- Energy transfer via magnetic mirror force
- Also known as "Barnes Damping", (Barnes, 1966)

NOT Intended Meaning:

- Earliest documented use found in Ishida & Nishikawa (1975)
- Relates to Langmuir wave packet collapse (Graham et al, 2012)
- Also called "localized Landu damping" (Robinson, 1989)
- Not a wave-particle resonant process (Skjæraasen et al , 1999)

Key Terms

- ***Transit-Time Damping (TTD):*** Magnetic analogy of Landau damping, where particles "surf" via magnetic mirror force.
- ***Plasma Turbulence:*** Multi-scale chaotic fluctuations in plasma flows and electromagnetic (EM) fields.
- ***Turbulence Damping:*** Collisionless decay of fluctuating bulk flows and EM fields, energizing particles.
- ***Field-Particle Correlation (FPC):*** A quantity (Klein & Howes, ApJL, 2016) that correlates fields and particle data from single-point measurements to reveal unique velocity space signatures of various turbulence damping mechanisms, usually denoted by $C(v_\parallel, v_\perp)$.

1. What is Transit-Time Damping?

Transit-Time Damping is:

- A collisionless damping process in plasma turbulence that ***flattens the particle distribution function.***
- Occurring in a time-varying field through interactions between resonant particles and magnetic mirror force, provided by ***the parallel gradient of magnetic field strength*** $\nabla_\parallel B$
- Energizing particles via ***the perpendicular component of the induced electric field*** $\vec{E}_\perp$

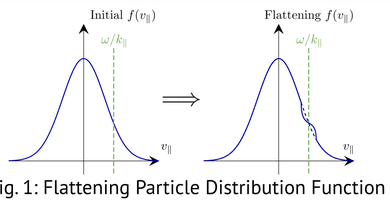

Fig. 1: Flattening Particle Distribution Function

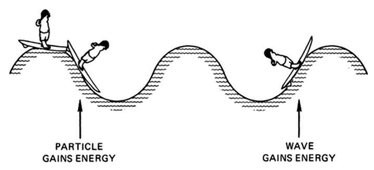

Fig. 2: Surfing Picture, (Chen, 2016)

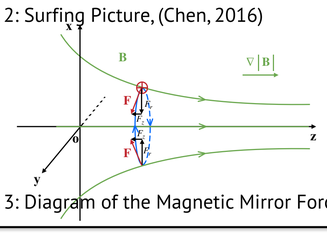

Fig. 3: Diagram of the Magnetic Mirror Force

But Exactly How do Particles Exchange Energy with EM fields?

$\Longrightarrow$ Check the $v_\perp$ vs $v_\parallel$ plot of single particle motion (SPM) in a mirror field.

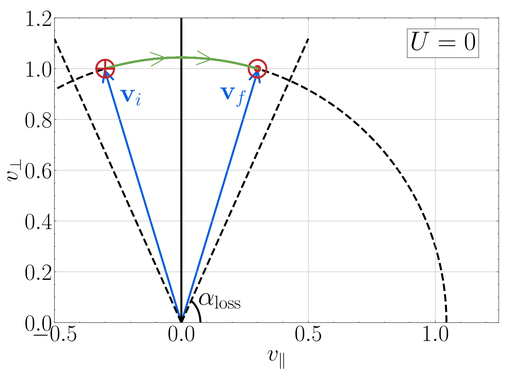

Fig. 4: SPM in a static field. Total kinetic energy remains unchanged.

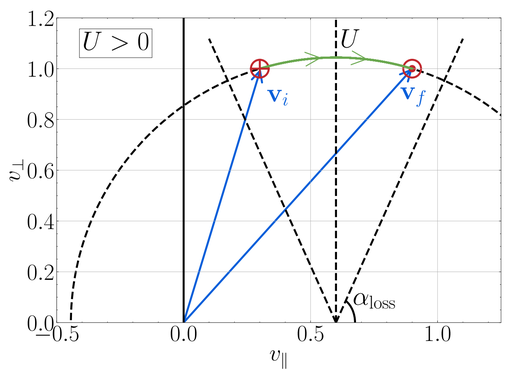

Fig. 5: SPM in a moving field with a speed of U. Total kinetic energy changes.

Comparisons with Landau Damping and Cyclotron Damping:

	Transit-Time Damping	Landau Damping	Cyclotron Damping
Resonant Condition	$\omega - k_\parallel v_\parallel = 0$	$\omega - k_\parallel v_\parallel = 0$	$\omega - k_\parallel v_\parallel = n\Omega_s, n \neq 0$
Energizing Particles by	$\vec{E}_\perp$	$E_\parallel$	$\vec{E}_\perp$

2. Method: Field-Particle Correlation (FPC)

Two different forms of the definition of TTD FPC:

Form 1:

$$C_{E_\perp,s}(\vec{r}_0, \vec{v}, t; \tau) \equiv -\frac{1}{\tau}\int_{t-\tau/2}^{t+\tau/2} \left[(E_{\perp 1})\left(q_s \frac{v_{\perp 1}^2}{2}\frac{\partial f_s}{\partial v_{\perp 1}}\right) + (E_{\perp 2})\left(q_s \frac{v_{\perp 2}^2}{2}\frac{\partial f_s}{\partial v_{\perp 2}}\right)\right] dt'$$

effectively a moving-average with a window size of τ, the correlation interval, to smooth data

Form 2:

$$C_{\delta B_\parallel,s}(\vec{r}_0, \vec{v}, t; \tau) \equiv \frac{1}{\tau}\int_{t-\tau/2}^{t+\tau/2} (\nabla_\parallel \delta B_\parallel)\left(m_s \frac{v_\parallel^2 v_\perp^2}{4B}\frac{\partial f_s}{\partial v_\parallel}\right) dt'$$

Compare to the Pearson product-moment correlation coefficient:

$$\rho_{X,Y} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}$$

1) no subtraction by mean value
2) no normalization by variance

Relations to Energy:

$$C(v_\parallel, v_\perp) = \left\langle \frac{\partial w_s}{\partial t} \right\rangle_{\text{Time Average}},$$

where $w_s \equiv \frac{1}{2} m_s v^2 f_s(\vec{r}, \vec{v}, t)$ is the defined ***phase-space energy density.***

Other quantities of interests:

- $W_s = \int w_s d^3\vec{v}$, energy density
- $\mathscr{W}_s = \int W_s d^3\vec{r}$, energy
- $C(v_\parallel) = \int C(v_\parallel, v_\perp) dv_\perp$, reduced correlation

3. Velocity Space Signature of TTD

AstroGK (Numata et al, 2010) Single Kinetic Alfvén Wave (KAW) Simulation Data,

$\beta_i = 1, k_\perp \rho_i = 1$

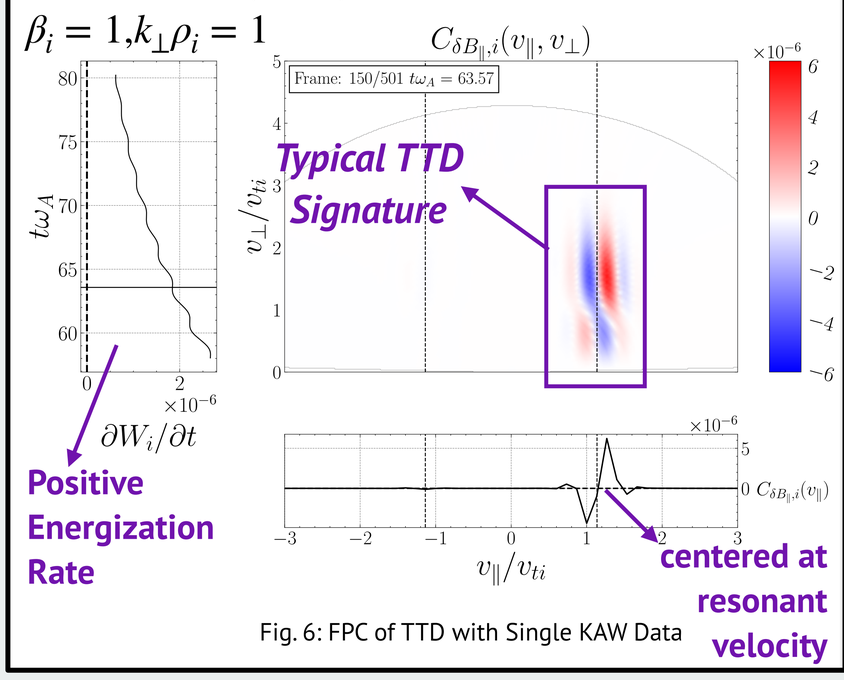

Fig. 6: FPC of TTD with Single KAW Data

Features:

- Bipolar pattern centered at the resonant velocity.
- Blue and red part represent loss and gain of phase-space energy density, respectively.
- Floats away from $v_\parallel$ axis, as expected from $C_{\delta B_\parallel}$ formula and the confinement of the loss cone angle.

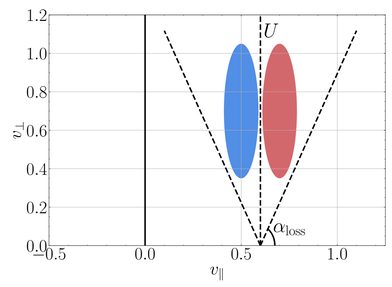

Fig. 7: Diagram of TTD Signature

4. How to Identify TTD From Plasma Turbulence?

Step 1: Obtain data.

$\Longrightarrow$ AstroGK Turbulence Simulation Data, $\beta_i = 1$, Four KAW Modes Initialized.

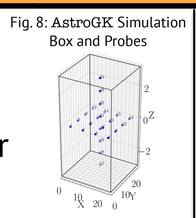

Fig. 8: AstroGK Simulation Box and Probes

Step 2: Choose the proper form of FPC.

$\Longrightarrow$ Since AstroGK evolves gyrokinetic equations and does not output $E_\perp$, choose $C_{\delta B_\parallel}(v_\parallel, v_\perp)$.

Step 3: Calculate the reduced correlation $C_{\delta B}(v_\parallel)$ and stack it along the time axis (we call it "timestack" plot) and look for time intervals where the bipolar pattern persists.

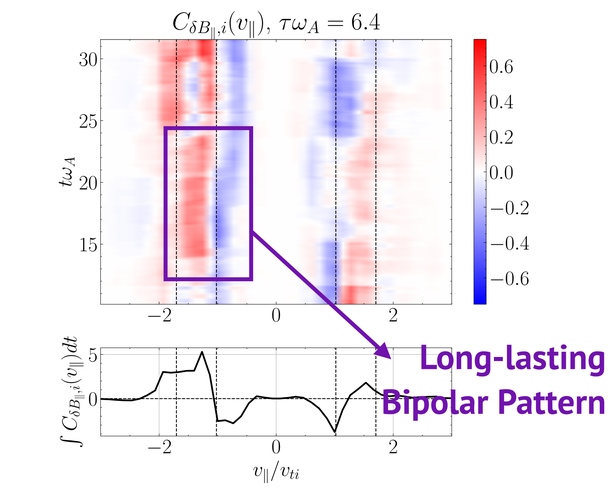

Fig. 9: Reduced FPC of TTD with Turbulence Data

Step 4: Plot the correlation $C_{\delta B}(v_\parallel, v_\perp)$ for each time frame within that time interval (preferably generate a video) and identify the floating bipolar pattern as the TTD signature.

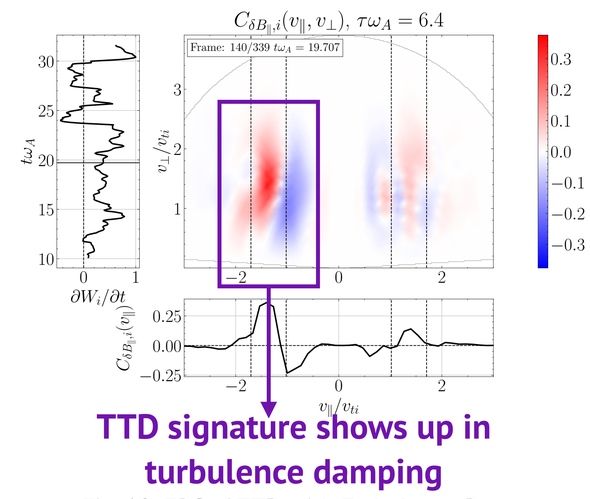

Fig. 10: FPC of TTD with Turbulence Data

This is ***the first numerical simulation showing*** that TTD contributes to the wave damping of low-frequency magnetized plasma turbulence.

Recommended Reading

- Huang, R., Howes, G. G., & McCubbin, A. J. (2024). The Velocity-Space Signature of Transit-Time Damping. arXiv preprint arXiv:2401.16697, Accepted in Journal of Plasma Physics.
- For more references, please scan the QR code on the right.

More Info: